J. Inst. Eng. India Ser. A (September 2024) 105(3):749–766
https://doi.org/10.1007/s40030-024-00831-3

REVIEW PAPER

Study on Life Cycle Assessment and Life Cycle Sustainability Assessment of Buildings: A Review

Manali Basu[1] · **Nagaraju Kaja**[1] · **Prashanti Rao**[1]

Received: 13 November 2023 / Accepted: 1 August 2024 / Published online: 5 August 2024
© The Institution of Engineers (India) 2024

Abstract As per United Nations (UN), fossil fuels contribute largely to climatic changes, and they are liable for emitting above 75% of greenhouse gases globally. The various other causes of climate change include heat and electricity generation, manufacturing of goods, deforestation, agriculture and changes in land use, transportation running on fossil fuels, production of food, consumption of goods such as plastics, electronics and clothing, and others. Further, the fast growth of the construction industry in the past few years increased the amount of greenhouse gas emissions all over the world. Commercial buildings and residences expend above half of the overall energy globally, also there has been a gradual rise in carbon dioxide emissions from buildings in current times. Therefore, weather changes and global warming have become global issues, due to which sustainable development has gained major prominence in the field of construction, to reduce damages to the environment. Life Cycle Assessment (LCA) is a tool which is useful for evaluating such concerns pertaining to environment on a 'cradle-to-grave' basis. Life Cycle Sustainability Assessment (LCSA) is deemed to be as LCA's future. This paper aims to furnish a review of LCA and LCSA studies performed by various researchers all over the world, to understand its present status with respect to different types of LCA and LCSA studies in the building and construction fields. This study's novelty lies in the fact that it has further aimed to find out whether culture, the fourth pillar of sustainability has been addressed in building LCA and LCSA studies in the past. The findings of this study have been summarized in the conclusion section which can be taken up by researchers for future studies.

Keywords Buildings · Life cycle assessment · Life cycle sustainability assessment · Literature review

Introduction

The construction industry occupies the second highest position among several other types of industries, when it comes to emission of greenhouse gases (GHG) during both the construction and operation phases [1, 2]. It is also a primary cause of resource depletion [2]. In the recent situation, when there is an urgent need to shift to sustainability in the construction industry, it has been recognised widely that the buildings are majorly consuming energies which are based on fossil fuels and emitting carbon dioxide [3]. In India, towns and cities have expanded due to urbanisation. This increased urban population requires more buildings to be built where people can live and work [4]. Currently, India occupies the fourth position for being the largest consumer of energy after countries like China, the USA and Russia. In India, residential buildings consume 24% out of 33% of the whole energy produced [5]. In order to create a future which is truly sustainable in all aspects for human as well as the Earth, the indicators that contribute to such sustainability need to be measured for any field or practice [4]. LCA can be used to evaluate and measure the emissions which are related to buildings [6].

Life Cycle Assessment has gained wide popularity because it treats the impact assessment, framework and data quality in an integrated manner [7]. ISO 2006 has divided LCA into four different phases, which include

✉ Manali Basu
basumanali94@gmail.com

[1] Department of Architecture, School of Planning and Architecture, Vijayawada, Andhra Pradesh 520008, India

defining the goal and scope, inventory analysis, impact assessment and interpretation [8]. ISO 14025 specifies 'Global-warming potential (GWP),' 'Acidification potential (AP),' 'Abiotic Depletion Potential (ADP),' 'Eutrophication Potential (EP),' 'Ozone Depletion Potential (ODP)' and 'Photochemical ozone creation potential (POCP)' as the primary environmental impact categories [1]. ISO 14040 forms the basis for LCA methodology [7]. Ciroth et al. [9] considered LCSA as LCA's future, because of its immense capability of representing a wide range of impacts, which is beyond the purview of conventional LCA, that primarily focuses on the impacts on the environment. In a review carried out by Dong et al. [10], the authors aimed to analyse the conformity of existing studies related to LCSA with Life Cycle Initiative's principles in the fields of building and construction. It was observed that the 'transparency (P8)' as well as 'completeness (P3)' principles got maximum average scores, while the 'key stakeholders (P4)' along with 'product utility (P5)' principles were the most challenging to comply with. Through a literature review, the present knowledge and methodologies have been analysed by Larsen et al. [11] for integrating LCA, Life Cycle Costing (LCC), and Social Life Cycle assessment (S-LCA) into LCSA, which will serve as a holistic approach, and will further help in the transition of the real estate and construction industries into a circular economy. The authors said that for supporting the transition process, further development in S-LCA is required, as it will be an important aspect for increasing attention on the use and recycle or reuse phases. In another review undertaken by Backes and Traverso [12], the authors defined how and whether LCSA is interpreted and applied in the construction field in the present day. It was found that out of the total number of papers that were reviewed, only 11% considered all the three main pillars of sustainability, namely social, environmental, and economic, for assessment purpose. 42% of studies used Net Present value (NPV) for LCC, while for performing S-LCA, major focus was given to the production stage.

In the studies discussed above, it is observed that there is a growing interest among the researchers to address all three pillars of sustainability and consider LCSA which is a more holistic approach. The previous authors have also emphasized on the fact that S-LCA requires further development. But apart from environmental, economic, and social, one more aspect is there, which is referred to as sustainability's fourth pillar, and that is culture [13], thereby forming the 'quadruple bottom line.' So, there is a need to conduct a literature review to understand whether culture has been taken into account in LCA or LCSA studies in the fields of building and construction. Therefore, this review's principal objective is to present the current state-of-the-art of various types of LCA and LCSA studies in the construction industry performed till date, by various researchers all over the world, to understand the lacunae and future scopes of study in this field. Further, the uniqueness of this study is that it has undertaken a literature review to understand whether culture has been addressed in past building LCA/LCSA studies, as addressing culture, which forms the heart of the 'Sustainable Development Goals (SDGs)' according to UNESCO, is significant to achieve cultural sustainability, which also plays a pivotal role in accomplishing SDG 11, especially target 11.4. The findings resulting from reviews and discussions in this study will furnish research directions for future researchers in this field who can consider the lacunae that are still existing to bridge the gaps in their studies.

Methodology

The screening of papers for review has been done systematically according to the 'Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)' guidelines [14]. Keywords like 'life cycle assessment,' 'life cycle sustainability assessment,' 'buildings' and 'construction industry' and the titles of the records have been used for identifying literature in 'Scopus,' 'Web of Science' and 'Google Scholar' databases. A total of 2093 records were identified which included mainly journal research articles. After that 92 duplicate records were removed, and 2001 records were screened further. These records were further reviewed by the abstract which resulted in the exclusion of 1672 records. 329 records were sought for retrieval, out of which 278 have been excluded as the full text was not available and they were not in the time range of 2012–2022, that is the years for which the review has been conducted. A total of 51 records have been then assessed by full text and no more records have been excluded. So, a total of 51 research articles, both at the national (India) and international levels, have been included in the review process. Apart from these 51 shortlisted research articles, six review articles at the international level and one review article at the national level along with eight case studies and four studies that have included culture in LCA and LCSA studies have also been reviewed in this paper.

Results

Different Stages, Types, and Methods of LCA

The LCA framework comprises of four different stages, namely (i) defining the goal and scope, (ii) inventory analysis, (iii) impact assessment and (iv) interpretation [15]. The first half of stage one includes defining the goal, the intended application of the assessment as well as the

audience, while defining the scope includes defining the functional unit, the system boundary and allocation procedures for the study. The next phase deals with the gathering of primary data from the process and collating those data with the secondary data acquired from databases like 'EcoInvent' or 'Inventory of Carbon and Energy (ICE).' Third stage deals with the selection of impact category, classification, characterisation, normalisation followed by grouping and weighting. And finally, the interpretation stage deals with the recognition of the important issues, evaluating the study's completeness, checking the sensitivity as well as consistency of the assessment, followed by stating the conclusions, limitations, and recommendations [16]. Also, LCA can be of two types namely (i) 'attributional life cycle assessment (ALCA)' and (ii) 'consequential life cycle assessment (CLCA).' ALCA helps in estimating the share of the global environmental burdens belonging to a certain product, whereas CLCA provides an estimation of the ways in which the environmental burdens get affected globally due to the production and use of the product [17].

Further, the three methods which can be used in LCA are (1) 'process analysis,' (2) 'input–output (I–O) analysis' and (3) 'hybrid analysis.' Method 1 is widely used but it has certain disadvantages because of the difficulty in the upstream needs of goods and services. Several researchers consider the second method which is input–output analysis to be more exhaustive as compared to the first method, as it considers the average data at a national level for each of the economic sectors. For the purpose of eliminating horizontal and downstream truncation, 'I-O based hybrid analysis' merges the process and I-O data into 'process-based hybrid analysis.' Process analysis is used for calculating direct inputs for a particular product on which the study is being carried out. The reliability of analysis is increased by using process data, but the required data is not easily obtainable. Each of these methods have their own advantages and disadvantages. Process-based LCA is more process specific and more detailed results are obtained, however, it becomes subjective when it comes to setting the system boundary, it is costly and time intensive, and there might be uncertainty in data. On the other hand, input–output based LCA allows systems-level comparisons and comprehensive assessments, however it becomes difficult when it comes to process assessment, there is a lack of data for entire effects on the environment, it is tough to implement to an economy that is open in nature, and there might be unreliability in data in this case as well [18, 19].

Review of Literature (International)

Table 1 shows the years of publication, locations of past studies, and various aspects that have been addressed by previous researchers in building LCA studies at the international level. A review of these studies has been presented in this section.

In E-LCA studies, Pachta and Giourou [20] found that the historic and modern school buildings, whose life spans were considered as 140 and 60 years respectively, presented different environmental performances and impacts when LCA was performed for cradle-to-grave and beyond building life stages, even though they had same operational characteristics and needs. The authors used 'Athena impact estimator for buildings' for carrying out the analysis. It was found that the values of 'Global Warming Potential (GWP)' and consumption of energy during the operational phase were higher in case of the historic building because of its increased life span and a lack of insulation in its structure, but the embodied energy was considerably lower as it was made up of natural building materials. Ryberg et al. [21] evaluated environmental impacts of four buildings (concrete, CLT (cross laminated timber), timber framed buildings and an existing concrete building that has been renovated) over 30 years of service life at the midpoint indicator in order to find out which among the selected four buildings had the smallest impacts on the environment on a cradle-to-grave basis. The study considered 1 m^2 of a dwelling as the functional unit. 'Ecoinvent 3.4 cut-off LCI database' and OpenLCA software were employed for performing the analyses. The CLT (cross laminated timber) building performed the best, and the concrete building performed the worst environmentally. Chen et al. [22] investigated environmental impacts of CLT and compared between timber and concrete buildings by taking into account two 8-story residential buildings on a cradle-to-gate basis for a service life of 80 years. One m^2 of the floor area was considered as the functional unit, and several databases in SimaPro, including USEI and Ecoinvent databases together with SimaPro v.9 software were used for the study. A reduction by 25% in 'Global Warming Potential' was observed for the building made of timber, which performed better environmentally in various impact categories as compared to the concrete building which was considered as the baseline case. In certain categories like AP (acidification potential) and SFP (formation potential of tropospheric ozone), the performance of the timber building was not well, which might be due to the greater distance of transportation that was needed for CLT (cross laminated timber). Jahandideh et al. [23] analysed and assessed carbon emissions and the carbon footprint of two concrete residential buildings for balancing carbon emissions during 50 years of service life of the structures on a cradle-to-grave basis. The authors said that through design optimization and customization of the constituents of concrete, and an improved energy consumption by the citizens shall significantly help in reducing carbon emissions from concrete structures. Lankhorst et al. [24] performed a comparative analysis of the impacts on the environment of different structural systems for office

Table 1 Year of publication, location of the studies and aspects considered by past researchers in LCA studies of buildings at the international level (Source: Author)

Sl. No	Source	Year of Publication	Location of the study	E-LCA	LCC	S-LCA	LCSA	C-LCA	ENG-LCA	BT-LCA
1	[20]	2022	Naoussa, N. Greece	✓						
2	[21]	2021	Greenland	✓						
3	[22]	2021	Chongqing, China	✓						
4	[23]	2020	Malaysia	✓						
5	[24]	2019	Rotterdam, Western Europe	✓						
6	[25]	2019	Dalarna, Sweden	✓						
7	[26]	2019	Korea	✓						
8	[27]	2017	Campania, South of Italy	✓						
9	[28]	2015	Vancouver (BC, Canada)	✓						
10	[29]	2013	Italy	✓						
11	[30]	2021	Sweden		✓					
12	[31]	2017	United States		✓					
13	[32]	2007	–		✓					
14	[33]	2022	Italy	✓	✓					
15	[34]	2021	Porto, Portugal	✓	✓					
16	[35]	2021	U.S. Pacific Northwest	✓	✓					
17	[36]	2021	Lisbon and Budapest	✓	✓					
18	[37]	2019	Borlänge, Sweden	✓	✓					
19	[18, 19]	2016	Gaziantep, Turkey	✓	✓					
20	[38]	2021	Slovenia							✓
21	[39]	2021	South Moravia (Czech Republic, Central Europe)							✓
22	[40]	2019	College Park, Maryland							✓
23	[41]	2019	Sweden							✓
24	[42]	2019	City of Gothenburg							✓
25	[43]	2019	Borlänge, Sweden							✓
26	[44]	2019	Alexandras Avenue, Athens							✓
27	[45]	2013	Case study sites at Paris/France, Alingsås/ Sweden, and Delft/the Netherlands							✓
28	[46]	2021	Greece						✓	
29	[47]	2018	Singapore			✓				
30	[48]	2014	The whole of Islamic Republic of Iran for analyzing the hot spot Northern parts of Iran, comprising of 3 provinces namely Golestan, Mazanderan and Gilan for site specific analysis			✓				
31	[49]	2022	–				✓			
32	[50]	2021	Tehran, the capital city of Iran				✓			

Table 1 (continued)

Sl. No	Source	Year of Publication	Location of the study	E-LCA	LCC	S-LCA	LCSA	C-LCA	ENG-LCA	BT-LCA
33	[51]	2018	Malaysia				✓			

E-LCA, Environmental-LCA; C-LCA, Cultural-LCA; ENG-LCA, Energy-LCA; BT-LCA, Building Transformation-LCA

buildings, which are high rises, having a range between 150 and 250 m on a cradle-to-gate basis, by considering the building's main load—bearing structure as the functional unit. The authors used 'Dutch National Environmental Database (NMD)' and 'Bouwen met Staal (BmS)' databases together with 'Grasshopper,' 'Karamba' and 'Excel' software for carrying out the analyses. It was found that frame and tube structures were responsible for the highest cradle-to-gate environmental impacts in terms of stability systems. The lowest environmental impacts were observed for the outrigger and diagrid structures for all heights. Compared to the steel models, the outrigger structure decreased the impacts by 5–16% for concrete and 20–26% for steel models. Petrovic et al. [25] assessed environmental impacts of materials of construction for a residence in Dalarna, Sweden, for a service life of 100 years on a cradle-to-grave basis, however, the installation phase, operational energy and water usage were excluded from the analysis. 'One Click LCA software' was employed for performing the assessment. It was found that the concrete slab and thermos—treated wood showed major contributions to carbon dioxide emissions, followed by wooden frames, whereas cellulose insulation had low environmental impacts. Almost all the environmental indicators showed environmental impacts which were quite high for production as well as replacement stages, whereas transportation along with deconstruction stages showed the least. The foundation contributed the most to the global warming potential (GWP). Roh et al. [26] estimated the embodied environmental cost during the life span of six main building materials, namely 'ready-mixed concrete (RMC),' 'glass,' 'concrete brick,' 'rebar,' 'expanded polystyrene and expanded polypropylene insulation' and 'gypsum board' of 443 Korean apartment buildings for a service life of 40 years on a cradle—to—grave basis. 'Monte Carlo simulation (MCS)' and 'KOLID methodology (Korean Life cycle Impact assessment method based on damage-oriented modelling)' were employed for performing the study. From the outcomes of sensitivity analysis, it was observed that RMC contributed the most to embodied environmental costs, followed by 'concrete brick,' 'rebar,' 'insulation,' 'glass,' and 'gypsum board.' The authors concluded that 'flat plate' or 'rigid frame construction' and 'tower type wall structure' can prove to be beneficial in achieving reduced embodied environmental costs during the life cycle. Vitale and Arena [27] aimed to develop a reliable and exhaustive inventory consisting of primary data which is of high-quality, and which can compare the impacts on the environment during the cradle-to-grave phases throughout the building life cycle. The authors considered a residential building for 50 years of service life for their study and considered the entire building's usable floor area as the functional unit. 'Ecoinvent 3.0.1' database along with 'SimaPro® 8.0.2' software was used for analysis. It was found that the largest contributor

was the use phase, contributing 77% to 'global warming' and 84% to energy consumption that is non-renewable. Concrete contributed to 26% of 'respiratory inorganics potential (RINP),' 40% of 'global warming (GWP)' and 25% of 'non-renewable energy (NREP).' Kumar et al. [28] evaluated and performed a comparative analysis of the impacts on the environment and consumption of primary energy of four types of residential buildings which consisted of two apartments, one a high-rise and the other low-rise, and two single family houses, one an attached house and the other a detached house for cradle-to-grave and beyond building life phases, with 1 m^2 of the floor area as a functional unit. The authors further performed a sensitivity analysis for selected houses over service lives of 50, 75 and 100 years respectively. DesignBuilder, Athena Impact estimator and @risk software tool were used for the assessment. The high-rise apartment performed the best and had least energy consumption and environmental impacts over its lifecycle, whereas energy requirements for the single-family detached house over its lifecycle were quite significant. Asdrubali et al. [29] discovered that the operational phase contributed very highly towards the entire impact, where for the detached house, it was 77%, and 85% in case of the building with an office over 50 years of service life by performing LCA for cradle-to-grave phases, and considering 1 m^2 of the usable or living floor area for a year as the functional unit. This was followed by the construction phase where the building containing the office had an impact of 14% and the detached house had an impact of 21%. 'Ecoinvent database' along with 'SimaPro 7.2' software were used for analysis.

Coming to Life Cycle Costing (LCC), Petrović et al. [30] in their study, found that a major part of the total LCC was represented by the construction costs, where half was for labour costs only, for 50 and 100 years of service life of a residence. The long-term cost of the dwelling was estimated for cradle-to-grave stages by using 'Eurostat' database along with 'One Click LCA software.' Noshadravan et al. [31] analysed the costs that can be expected from damage caused by natural hazards, mainly earthquake and hurricane, by taking a case study of a wood-framed single-family townhouse for cradle-to-grave phases by using residential cost data database and 'BEopt simulation tool.' It was observed that repairing costs for damages caused by the hurricane can be noteworthy in comparison to other costs during the dwelling's life cycle. It was found that lifecycle costs are dominated by initial construction costs which is followed by energy, maintenance, and improvement costs. But, in some cases the hurricane repair cost can surpass all such costs, which includes the construction cost at the initial stage as well. Also, results are specifically dependent on the building's location, the way in which it has been designed, the mitigation measures that have been incorporated and the discount rates that have been implemented.

In the combined E-LCA and LCC studies, Vela et al. [33] found that the most significant costs were represented by initial costs, which was almost 60% of overall costs when a 50 years life span scenario was considered for a building with an office, for cradle-to-grave phases by using 'ISTAT' database. Carvalho et al. [34] aimed at developing a tool for designers which is based on BIM for making decisions, for evaluating the environmental, functional, as well as economic performance of various solutions for construction of buildings. The authors used Autodesk Revit for creating 18 types of different simulation scenarios with different types of external walls, floors, and roof combinations, and applied it to a detached single-family one-level residence as a case study for 50 years of service life with 1 m^2 of construction as the functional unit for cradle-to-grave phases. Further, the authors aimed at developing a framework in Dynamo for characterizing the impacts on the environment and the building elements' costs throughout their life span automatically, including automation of LCA as well as LCC assessment at the time of initial project stages. Liang et al. [35] examined the environmental and economic contributions of a building having mixed use (commercial along with residential) for 60 years of service life on a cradle-to-grave basis by considering 1 m^2 of living/working floor area as the functional unit. 'Athena Construction Energy Database,' 'DATASMART,' 'Athena Impact Estimator (AIE)' and 'RSMeans' databases along with SimaPro software were used for conducting the analysis. It was found that the operating energy contributed to over 90% of the greenhouse gas emissions from the building. Kiss et al. [36] developed an automated optimisation framework and then the authors applied the same to a residence which was taken up as a case study, by considering the entire structure as the functional unit for 50 years of study period and considered the cradle-to-grave phases. 'Ecoinvent v3.6 cut-off database' along with DesignBuilder and EnergyPlus, OpenLCA, Jupyter with Pandas and Matplotlib, Python, Monte Carlo software were used for the study. The authors concluded that to successfully find solutions for reduced environmental impacts and a thermal envelope which is eco-efficient, multi-objective optimisation algorithms can be used. Ramírez-Villegas et al. [37] did not observe any correlation between economic and environmental performances of a three-story structure with multiple dwellings consisting of 36 apartments for 50 years of service life, due to renovation strategies that were investigated in a Nordic climate. The authors took into consideration the whole structure as a functional unit and performed the analysis for cradle-to-grave phases by using 'International Labor Organization' database along with IDA-ICE, One Click LCA software. Atmaca [19] found that in both the contemporary residential buildings, one in urban and the other in a rural area, the operation phase dominated and contributed to 85–87%

of the main energy requirements and 82–88% of carbon dioxide emissions, for 50 years of service life for cradle-to-grave phases using ICE data database. Further, the authors found that life-cycle costs consisted of mortgage, energy, maintenance, service, and demolition phases.

Several researchers have addressed building transformations in LCA studies. Obrecht et al. [38] developed a new methodology to calculate residual value and environmental impacts of refurbishment measures in residential building by creating a division between life cycles for 'end of life (EoL)' and after the life phases of the building and used Pravilnik o vzdrˇzevanju stavb (used in Slovenia), Bewertungsystem für Nachhaltinges Bauen (used in Germany) and SIA2032:2020 (used in Switzerland) databases. Struhala and Ostrý [39] aimed to come up with new data on environmental impacts related to renovations in residential buildings for 60 years of service life by using LCA. The authors considered the cradle-to-grave phases with a functional unit of one m^2 of the treated floor area and used 'Eco-Bat 4.0' database with 'Eco-Bat 4.0' which is a Swiss software for performing LCA for buildings for the study. The results showed that for achieving sustainability in buildings, they should be renovated and energy sources that are environmentally friendly should be installed. Hu [40] aimed to create a model for LCA and test it on an education building with 40 years of service life for cradle-to-grave phases by using EIA 2018 database with 'Building Life Cycle Cost (BLCC) version 5.3-18,' Autodesk Revit, Sefaira and EnergyPlus software. It was found that the main factor in an existing building's LCC is the initial investment required for energy efficient renovations. Nydahl et al. [41] aimed to introduce a LCA method for building refurbishment, and further aimed to analyse a residential building as case study with the introduced method for cradle-to-grave phases by using IDA-ICE software. It was found that energy recovery ventilation was the most advantageous refurbishment measure. Österbring et al. [42] used explorative LCA for accounting for environmental impacts resulting from prevalent renovation measures, as well as the energy savings that would result till 2050, by considering a residential building for A1-A3 as well as B6 stages of life cycle according to Standards, by employing the Ecoinvent V3.1 database and SimaPro V8.0.5.13 software. The results showed the possibility of energy savings by almost 23% along with a minimization of 31% in the emission of greenhouse gases. Ramírez-Villegas et al. [43] aimed to find out the consequences of different types of renovation measures on the impacts on the environment of a residence during 50 years of its service life and for cradle-to-grave phases, for construction materials and consumption of energy during the operational period. The entire residence was considered as the functional unit and Ecoinvent and GaBi databases along with 'IDA-ICE (version 4.7.1. EQUA Simulation AB, Solna, Sweden)' software were used for the study. A significant impact on energy usage was observed because of the renovation measures, attributable to cold climatic conditions along with low solar irradiation during the heating season. Karoglou et al. [44] aimed to make provision for interventions, in order to enhance the sustainability conditions of a historic area comprising of refugee buildings, consisting of eight block apartment flats from the interwar period, by evaluating the environmental performance of the buildings whose conditions have worsened, and which serves as a landmark for the neighbourhood. A block comprising of six apartments with each apartment having 50 m^2 as the gross floor area, and having a common entry was considered as the functional unit. The GaBi database was used for the study. The authors further aimed to oppose the pulling down of the identified buildings from the twentieth century and considered preserving the modern built heritage from a perspective of wider regeneration and renewal. The study showed that a successful governing of the rehabilitation process of built heritage, by means of sustainability principles, can lead to a nexus of preserving the culture and sustainability. Ostermeyer et al. [45] in their study inferred that to respond appropriately to the current challenges and to come up with a holistic picture, development of social indicators must be strengthened in the building sector. The authors considered residential buildings for 30 years of service life and used the Ecoinvent 2.2 database along with the Passivehouse project tool (PHPP) for the study.

In an energy-LCA study performed by Dascalaki et al. [46], the authors compared the primary energy consumption of four single-family Hellenic houses with a service life of 70 years for cradle-to-gate phases by using the EcoInvent database with SimaPro and TEE-KENAK software. They also estimated the construction materials' embodied energy (EE) and aimed to find out the material which contributed remarkably to the overall embodied energy (EE). Concrete's contribution to the overall embodied energy was quite noteworthy, equalling to about 28%, which was followed closely by steel rebar, equalling to 26% among the four dwellings studied. Among the few studies that addressed the social aspects in LCA (S-LCA) of buildings is a study by Liu and Qian [47], who created a theoretical framework for the assessment of social sustainability by identifying relevant stakeholders and the associated impact subcategories with each of them for cradle-to-service phases. They applied the developed framework to two case study buildings, one a residential hall project for students, having a 'prefabricated prefinished volumetric construction (PPVC),' while the other a general residence that was a part of a previously published case study having a 'semi-prefabricated reinforced concrete structure.' For integrating the 'Social performance scores (SPS),' weights were generated among impact categories

as well as different phases of the life cycle. Three types of indicators, namely quantitative for generic analysis, and quantitative as well as indicators that are semi-quantitative in nature were considered for site-specific analysis, and their scoring and normalisation were done to a range of −2 to +2 for further integrating them into single social performance scores. It was found that the single social performance score for PPVC project was 0.926, which is why from the perspective of the life cycle, it had a collectively better social performance than the semi-prefabrication project whose single social performance score was 0.821. Hosseinijou et al. [48] compared the social as well as socio-economic impacts in the life cycle of concrete and steel as construction materials, for a dwelling and working place for 50 years of service life, by taking the quantity of concrete and steel required for 1 m^2 of the floor area as the functional unit and analysed the hotspot. For hotspot analysis, cradle-to-grave phases were considered, and the results of hotspot analysis determined the phases to be considered for the site-specific survey. Case study results showed that steel/iron performed better socially than concrete/cement.

The other LCA studies include a study done by Fajilla et al. [52], where the researchers evaluated the effects of behaviour of the occupants and change in climate on the environmental impacts caused by cooling along with heating systems in a dwelling in Southern Italy. It was observed that behaviour of the occupants influenced the impact of the systems quite greatly. Souza et al. [53] performed a comparative analysis of the impacts of coverings for the roof made of ceramic and also from concrete throughout their life span over one m^2 in Brazil. Results showed that tiles made of ceramic have considerably lesser impact as compared to tiles made of concrete on parameters like climate change, depletion of resources and withdrawal of water, however, in the other categories like human health and ecosystems quality, both the materials had very little differences, and hence were not considered to be significant.

Studies that addressed all three pillars of sustainability include a study done by Onososen et al. [49], where the researchers examined the important drivers of a 'Building Information Modelling (BIM)' based LCSA of a building. The study concluded that the essential drivers are organizational, governmental, and institutional support as well as capacity development. Oladazimi et al. [50] compared the sustainability of concrete and steel construction frames from a triple bottom line (TBL) approach for residential building on a cradle-to-grave basis. The total amount of material used in steel and concrete construction frames for 50 years of service life was considered as the functional unit and GaBi software was used for the study. It was observed that as compared to the steel frame, the concrete frame was responsible for almost 38% more environmental pollution. From LCC's perspective, the rate of the frame made of steel was almost 152,000 USD more as compared to the frame made of concrete. Concrete showed a little better performance socially, based on the opinions of expert, as the concrete buildings had a social dimension score of 0.199, whereas buildings made of steel had a score of 0.189. Balasbaneh et al. [51] analysed the environmental, economic as well as social impacts of various hybrid timber building constructions for residential building over 50 years of service life on a cradle-to-grave basis. The whole structure was considered as the functional unit and Refs. and Malaysia LCI Database along with 'SimaPro v.8.0.4.30 modelling' software were used for the study. The results showed that the timber sector shall not only provide higher wages to the existing employees, but also lead to increased job opportunities.

Also, a study of review papers has been undertaken apart from the research articles presented in Table 1. Features and shortcomings of three Building Information Modelling (BIM) software and eight tools for LCA were summarized in a review by Chen et al. [54]. The authors demonstrated that BIM software have excellent storage capabilities for building information, but also have issues such as data interactivity which include limitations in the type of file output. The authors determined that models with a lower level of development (LOD) are appropriate for early design phases. There are still problems in processing of dynamic data, so evaluation with present dynamic BIM integrated LCA tool should be integrated with 'Internet of things (IoT) big data information platform' and the database should be broadened as well. Also, integrating recent technologies for example, 'semantic web' and GIS will be helpful for improving technical performance as well as enhancing the efficiency of application. The study of review papers showed that limited consideration has been given to the influence of occupant behaviours within LCA modelling [55]. There are many LCA studies in the construction industry for assessing environmental impacts, reducing emissions, and for decision support, but Scherz et al. [56] raised the question of whether in the process of public building tendering and awarding also LCA is implemented or not, and further tried to identify the problems of implementing LCA in these early phases of a project. The authors found that several articles have talked about green public procurement or tendering that is sustainable in the field of construction, but, LCA has been implemented in very few procurement processes that is in only 19 articles out of the final sample that they analysed. The obstacles that were identified by the authors include difficulties with methodological, legal, organizational, economic and political aspects. The authors concluded that in future it should be mandatory to amalgamate LCA in the process of procurement, which will be essential for bringing down the carbon dioxide emissions and which will contribute to the climate target plan of the European Union

(EU) of ensuring carbon neutrality by 2050. Yildiz et al. [57] inferred that there is an effect of service life on the economic as well as environmental sustainability during the life cycle. In a review presented by Nwodo and Anumba [58], the authors said that to improve spatial and temporal accuracies of traditional LCA, dynamic building LCA is a promising approach and to resolve their challenges proper attempts should be taken. Janjua et al. [59] inferred that "carbon footprint" as an indicator is often used in assessing a structure's sustainability. The authors further inferred that there is a need for a framework which is comprehensive to address the 'Triple Bottom Line (TBL)' aspects of sustainability throughout the whole life cycle of buildings, as out of the 807 publications that they referred for their study, S-LCA and LCSA were addressed by only three and five studies respectively for residential buildings, while almost 80.4% discussed about E-LCA and 18.6% discussed both the economic and environmental aspects.

Review of Literature (National)

Table 2 shows the years of publication, locations of past studies, and various aspects that have been addressed by previous researchers in building LCA studies at the national level (India). A review of these studies has been presented in this section.

As compared to the international level, less studies on building LCA have been performed nationally. In E-LCA studies, Jain and Rawal [6] evaluated the gap existing between 'net-zero energy' and 'net-zero carbon' by means of LCA of a NZEB in the campus of CEPT University, for cradle-to-grave and beyond building life stages, however stages B7 (operational water use), C4 (disposal), D1 (reuse) and D2 (recovery) were excluded from the study. It was found that the building showed an impact which was negative with 866 tCO2 over 60 years of service life, although it has got the status of an annual net-zero operation building. Alotaibi et al. [60] proposed a method for LCA of a residential building during construction, operation, and demolition phases for analysing the embodied carbon over 75 years of service life. The authors used the ICE 3.0 database with Revit and One-click LCA software for analysis. It was found that carbon emission from the building was 414 kg CO_2e/m^2/year, also various decarbonization measures can minimize carbon emissions by 135 kg CO_2e/m^2/year when the initial analysis was considered as the baseline case. Kurian et al. [61] found that among the other materials studied in an apartment complex during the cradle-to-grave phases, cement contributed to carbon emissions to a major extent, and the highest amount of carbon was emitted during the operational stage, over 80 years of service life. The authors used GaBi, Ecoinvent and ICE (Inventory of Carbon and Energy) databases along with AutoCAD, Autodesk Revit Architecture 2018 and One Click LCA software for the study. In a study by Ameen [62] the author found that during the construction stage of a three-story commercial structure with a semi-basement, steel, cement, and brickwork mainly contributed to the GHG emissions whereas, tiles and glass played a very small role. Ishaq et al. [63] in their study found that the topmost consumer of energy was the operational phase that added more than 80% to greenhouse gas (GHG) emissions, while the second highest consumer was the construction phase, for an office and residence over 50 years of service life. The analysis was done for construction, operation, and maintenance phases by considering m^2 usable floor area as the functional unit with the help of AutoCAD software. Husain and Prakash [64] assessed the environmental impacts of a Government Polytechnic College by combining LCA with the Ecological Footprint indicator. It was found that when the annual grid electricity consumption was replaced by rooftop photovoltaic (PV) system connected to the grid in the ratios 100, 75, 50 and 25 percent, the total lifecycle ecological footprint of the building could be reduced by 61, 54, 31 and 15 percent respectively. Kumbhar et al. [65] identified and quantified the environmental performance of bricks by considering 16 traditional brick kilns for cradle-to-gate stages by using the Ecoinvent database along with SIMAPRO 7.3.3 software. It was found that sulphur oxides and particulates in traditional kilns served as the most offending pollutants, and they can be decreased by the total burning of coal which will enhance coal's efficiency as well as lower the problems related to emissions. Varun et al. [66] aimed to find out the greenhouse gas emissions and energy consumed by the building of the Mechanical Engineering Department at the National Institute of Technology in Hamirpur over 50 years of service life through LCA during the construction, operation, and maintenance phases by using m^2 usable floor area as the functional unit. It was found that the highest contributor to greenhouse gas emissions for all the three floors of the building were the framework made of reinforced cement concrete (RCC) and steel. Also, it was found that during the building's operation phase, 59% of the total energy was consumed.

Coming to LCC, Alone [67] aimed to estimate the costs of green and normal residential building constructions by using Revit Architecture, Revit MEP and Microsoft Excel. The author inferred that the cost of green building construction was about 20% more as compared to the normal one due to the cost of materials. Pawar and Marathe [68] in their study said that instead of the total initial constructional cost, the life cycle cost gives value perspective which is better than considering only the initial investment. The authors considered a residential building over 10 years of service life and employed the 'Net Present Value (NPV)' method for study. Kaur et al. [69] found that during the construction,

Table 2 Year of publication, location of the studies and aspects considered by past researchers in LCA studies of buildings at the national level (India) (Source: Author)

Sl. No	References	Year of Publication	Location of the study	E-LCA	LCC	S-LCA	LCSA	C-LCA	ENG- LCA	BT-LCA
1	[6]	2022	Ahmedabad, Gujarat	✓						
2	[60]	2022	India	✓						
3	[61]	2021	Ernakulam, Kerala	✓						
4	[62]	2019	Phagwara, Punjab	✓						
5	[63]	2019	Uttar Pradesh	✓						
6	[64]	2018	Madhya Pradesh	✓						
7	[65]	2014	Rural and urban parts of Palus block in Sangli and Karad in Satara district respectively	✓						
8	[66]	2012	Hamirpur, Himachal Pradesh	✓						
9	[67]	2020	Pune		✓					
10	[68]	2020	West Pune Zone		✓					
11	[69]	2020	India		✓					
12	[70]	2019	Gangapur rd, Nasik		✓					
13	[71]	2016	Pune		✓					
14	[72]	2022	Vijayawada						✓	
15	[73]	2012	Hyderabad						✓	
16	[74]	2021	Roorkee, Kurukshetra, Jammu (J1), Jammu (J2) and Ambala	✓					✓	
17	[75]	2020	Defence Colony, New Delhi	✓	✓				✓	
18	[76]	2021	–				✓			

operation and maintenance phases, brick wall costs less than autoclaved aerated concrete (AAC) and reinforced cement concrete (RCC) walls when the construction cost was considered, but when the operational cost was considered, reinforced cement concrete and autoclaved aerated concrete walls costed less than brick wall. The authors considered a building with an office over 50 years of service life. In a study by Deore and Ambre [70] the authors found that the costs required in the initial, operation and maintenance and replacement phases, and the non-annually recurring costs are the costs that are involved in LCC for a residence over a service life of 80 years when the construction, operation and maintenance phases were considered for analysis. Also, the Net Present Value (NPV) method helps in carrying out the LCC of projects that have a long-life span. The authors used data from the State Bank of India website and from the 'Ministry of Statistics and Programme Implementation Central Statistics Office, Government of India' for the study. Kale et al. [71] calculated LCC of two educational buildings over a service life of 25 and 30 years respectively by using Net Present Value (NPV) method and obtained necessary data from the Reserve Bank of India. By proposing energy efficient approach (EFA), it was found that implementation of solar panels can help in lowering the LCC of existing structures to a large extent.

In an energy-LCA carried out by Ganugula et al. [72], the authors inferred that the life cycle cost of a building which is energy efficient was 18.67% less in total as compared to a typical building over a lifespan of 30 years. The authors considered a residential building for the study and used Autodesk green building simulation, Revit Architecture and Revit Energy Modelling for analysis. T. et al. [73] considered a residential building as a benchmark building to understand its energy performance, and further modelled the selected building for five different climatic zones in India with different types of envelopes. Using the DesignBuilder database along with DesignBuilder software for analysis, it was found that most of the savings in lifecycle energy is achievable by adding insulation to external roofs and walls. Maximum savings in lifecycle energy due to insulation was observed for the warm and humid climatic zone whereas it was the least for the moderate climatic zone.

In a combined environmental and energy LCA study, Kumar et al. [74] said that reduced life cycle energy can be achieved by using materials having low energy, and appliances which are rated with low-energy-star and lighting fixtures. The authors considered a residential building for the study over 100 years of service life for cradle-to-grave phases and used MATLAB for analysis. In a study by Shrikhande and Desale [75], the authors said that the operational phase served as the most dominating phase over a service life of 50 years of an office building when analysis was done for cradle-to-grave stages. The authors used GRIHA Manual 4 as the database along with Conditional Demand Analysis and E-QUEST software for analysis. In an LCSA study, Srivastava et al. [76] inferred that the interactions among the social, environmental, and economic aspects vary while moving from the one phase to a different phase of the life cycle. The authors developed a methodological framework which has a provision for customization in order to meet the requirements for assessing sustainability of various phases and construction types in the future. In a review done by Yadav et al. [77], the authors found that generally 10–20% of the energy during the course of the life span in total is embodied energy while the rest 80–90% is the operational energy, and timber is preferable over concrete for building construction.

Review of Literature (Case studies)

This section deals with the review of case studies undertaken in the building LCA field. For understanding the main factors contributing to environmental impacts and embodied carbon of high-rise structures constructed with reinforced concrete, Ma et al. [78] took a 60 story high-rise building which has an office in Chicago, USA as a case study which has a steel composite and a core outrigger-reinforced concrete structure. The authors used 'building information modelling (BIM)' based LCA in Tally as well as Revit along with GaBi Database life cycle inventory (LCI). The structure's entire lifecycle stages as well as beyond life stages were considered, however, the use stage was excluded as the principal focus of research, which comprised of the structure of the building, did not have the requirement of any maintenance, repair or replacement throughout its normal lifespan. The results of this case study demonstrated that it is significant to lower concrete's embodied carbon in buildings. These results will give valuable insights to architects as well as engineers for understanding the main factors that contribute to environmental impacts and embodied carbon of high-rise structures. Kirilmaz and Metin [79] took Sumerbank Kayseri Textile Factory, which is at present a part of the main campus of Abdullah Gul University in Turkey as the case study building to select an appropriate wall panel for internal wall during its restoration phase, which will have less effect on the environment by comparing the environmental impacts of three different wall panels namely reinforced gypsum, gypsum and cement-based panels. The authors considered the cradle-to-grave phases for the study, obtained necessary data from market analyses and the EcoInvent database and used SimaPro software for the analyses. It was found that the most harmful wall panel was reinforced gypsum panel while the least harmful was the gypsum panel. Lachat et al. [80] considered two demolition sites as case studies that included residential building A, in the southern part of France and residence B, in the northwest part of France. The objective

was to develop an inventory for life cycle comprising of primary as well as secondary data related to the destruction of the selected buildings, to quantify potential environmental impacts that will originate from their end of life, and during the manufacture of aggregates that are recycled by using waste generated during construction and demolition, and to make a comparison between the impacts on the environment resulting from the manufacture of recycled aggregates and virgin ones. Ecoinvent® database version 3.5 along with OpenLCA v.1.10 software and impact method from the 'EN 15804 + A1' standard was employed for the analyses. This case study bridged the gap in literature by quantitatively evaluating the impacts on the environment related to aggregate that is recycled produced from a building's construction and demolition waste at the end of its life. Recycled aggregates showed an impact which was higher in comparison to the virgin ones for almost every indicator, excluding 'abiotic resources' and 'ozone depletion,' which was not analysed for aggregate that is virgin in nature. It was further inferred that sustainability effects of aggregates that are recycled, under the concept of circular economy, is highly sensitive to context, location, and cases. Marrero et al. [81] took 87 industrial buildings as case studies in three municipalities of Los Alcores, namely, Alcalá, Carmona and Mairena del Alcor in Seville, Spain to put forward a method for evaluating small scale construction projects belonging to the typology of industrial buildings for introducing solutions that are eco-efficient in nature. A structured method was proposed for evaluating construction of industrial buildings. It was inferred that the material that is consumed most does not always have more impact, like aggregates and stones in case of carbon footprint, whereas materials that were consumed less, like the plastics family showed high water footprint. Most concerning materials in both carbon and water footprint were concrete and cement, while the second place was taken by metal and alloy families.

Morsi et al. [82] took a residential building in the city of New Cairo in Egypt which was under construction as case study to discuss the impacts on the environment of the structure for cradle-to-cradle stages for a projected period of 60 years of life span. The authors used Revit and Tally environmental impact Plug-in for analysis. It was found that energy used in the operating phase contributed to more than 50% of the entire environmental impacts, where a share of about 60 to 76% was due to structural elements made of concrete. It was followed by the manufacturing phase which contributed to about 18–34% of the total impacts. The most critical construction material was concrete according to sensitivity analysis. Tulevech et al. [83] considered a low-energy structure belonging to the industrial typology in Thailand which was under construction as case study for a life span of 20 years. The objectives of the study were to simulate the demand of energy which is primary for every life cycle stage defined by focusing on the relationship between operational and embodied energy which is dynamic in nature, to compare the outcomes with those of institutional and commercial buildings from literature, for modelling the environmental burdens of each phase during the life cycle and to make a comparison between the results obtained from base case with three scenarios created. Inventory data for the study was obtained from the Ecoinvent version 3 database and SimaPro 8 software was used for analysis. This case study went beyond the energy savings initiative restricted to only commercial and residential buildings to focus on a cleaner industrial building sector, especially in the emerging nations with warm climates. The outcomes of the study showed that integrating a base case that is designed intelligently along with recycling of material and rooftop photovoltaic system can convert a 15,000 m^2 industrial building into a net-zero one. Hong et al. [84] took a green building which has an office in Guangdong Province, China as case study, to understand the accomplishments of technologies related to green building by finding energy performance of the structure during its life cycle, and to identify main points in construction and operation of green buildings. It was inferred that improvement can be achieved by not only controlling the extensive use of materials like concrete during the construction stage, but also by reducing the application of materials having energy intensity which is high, such as aluminium and steel. The results of this case study can act as a reference for improvements in green building development in the future. Collinge et al. [85] took the Mascaro Center for Sustainable Innovation (MCSI) building, a green building at the University of Pittsburgh in the United States as case study. The objectives were to monitor the indoor environmental quality and consumption of energy in the selected building, and to incorporate the measured data into a life cycle assessment which is dynamic in nature. It was concluded that dynamic LCA can show a marked difference in building evaluations because of their long lifespan and potential of undergoing changes. Integrating building monitoring at the post-construction phase with dynamic LCA modelling can help in improving accuracy while estimating building impacts.

Review of Literature (Culture in LCA studies of Buildings)

Apart from the three pillars of sustainability, culture is nowadays being considered as the fourth pillar [86]. The concept of culture in LCA has been reviewed separately as very few studies have addressed it. Pizzirani et al. [87] aimed to distinctly explore the potential of representing and including cultural aspects in LCSA along with environmental, economic, and social aspects. So, the authors conducted a study for investigating alternative options for

forestry in the Ngāti Porou land of the native people of New Zealand known as the Māori. The research suggested that there is a scope for culture to be included in LCSA, and representing culture distinctively from the social aspects might prove to be advantageous, especially if the final users have precise cultural needs and concerns. Pizzirani et al. [88] observed that addressing cultural values would have some potential benefits, and future studies should include culture into the LCSA process. Also, inclusion of culture in LCSA might help in protecting the rich cultural diversity found in different communities worldwide. Pizzirani et al. [87] said that cultural indicators, in spite of their presence in the subcategories of S-LCA, are generally less, and are not represented properly in the S-LCA studies. The authors further said that placing cultural indicators within the social impact categories limit their visibility, as well as imply that culture is nothing but a construct of social systems. As observed in S-LCA methodology, the amalgamation of cultural and social indicators does not help the cultural indicators to have their visibility separately and distinctly. Ketola and Salmi [89] conducted a holistic LCSA that included culture to compare different biofuels. From the results of the interviews related to culture, it was found that many of the stakeholders felt that cultural landscape will be lost due to the growing stages of the crops that are used for producing biodiesel. Weidema and Lindeijer [90] recognised culture in LCA with the aim of contributing to the evolution of 'missing' elements of LCA which included 'cultural values.' The authors interpreted the 'cultural values' for archaeological sites and landscapes. However, from Tables 1 and 2, it is observed that no such studies are there that has addressed the cultural aspects exclusively in LCA or LCSA study of buildings.

Discussion

From the review of literature at the international and national levels, it is observed that when it comes to the functional unit, most of the researchers have considered either 1 m^2 of the floor area or the entire building for their studies. Similar observation has been presented by Soust-Verdaguer et al. [91]. However, in case of India, it is observed that only few researchers have defined a functional unit for their studies. This issue has also been addressed by Nwodo and Anumba [58] in their study, where the authors said that studies related to building LCA have not defined functional units clearly. Almost 51% of the studies reviewed in this paper have considered the system boundary as the cradle-to-grave for conducting LCA. This finding aligns with Rashid and Yusoff [92] who said that the system boundary that is more often adopted for building LCA studies is cradle-to-grave. Three studies at the international level have also considered the stages of the life cycle beyond the building life. In case of studies from India, some researchers have considered only the construction, operation or use and maintenance phases for their study. Nearly 24% of the reviewed articles have not defined the system boundary in their studies. Rashid and Yusoff [92] and Mastrucci et al. [93] said that system boundary should be defined transparently. The authors further said that there is uncertainty in the ways in which the system boundaries are selected for building LCA.

Coming to the databases used by past researchers, almost 24% have referred to the Ecoinvent database while few have referred to GaBi and other databases. In India, some researchers have derived necessary data from several bank websites such as from the State Bank of India and the Reserve Bank of India. Almost 16% of studies have used SimaPro software, whereas 9% of studies have used One Click LCA software, 4% have used OpenLCA and another 4% have used Athena impact estimator for buildings software. Also, an increasing trend has been observed in the integration of LCA with BIM by using software such as Autodesk Revit. Almost seven, i.e. about 14% of the total number of studies reviewed in this paper have used Autodesk Revit. Especially, in India Revit has been used by four out of these seven studies. This observation aligns with a study's findings done by Fnais et al. [55], where 20% of studies analysed by the authors used BIM. Also, an identical finding has been presented by Obrecht et al. [94] in their study. Integrating BIM with LCA is necessary for simplifying the application of LCA to buildings as it will help in reducing the effort and time required for managing LCA data and information related to the building being assessed [95]. However, the authors pointed out that there are certain limitations of using BIM for LCA as the environment of BIM lacks certain aspects which are critical, and which are significant for assessing the impact on the environment as well.

Coming to the typology of buildings considered by past researchers, it is observed that almost 67% of studies reviewed in this paper have performed LCA for residential buildings. Most of the studies have considered the buildings' service life as 50 years which is about 35% of studies reviewed in this paper, whereas 6% have considered 80 years and another 6% have considered 100 years. While the others have considered different service lives for the buildings. Addressing this aspect, Nwodo and Anumba [58] said that it will be more realistic if the life span or service life of the buildings are assumed based on a building that is being considered for case study instead of assuming based on common practices.

Now coming to aspects addressed by past researchers in building LCA studies, it is observed that most of the studies, that is 18 among the 51 articles (35%) reviewed in this paper have considered only the environmental aspect,

eight out of 51 studies (16%) have considered only the economical aspect, whereas only two studies out of 51 (4%) at the international level have addressed sustainability's third pillar, namely, social. The Triple Bottom Line (TBL) approach considering all the three pillars of sustainability has been addressed by only four studies (8%), three at the international level, while only one at the national level. 12% of reviewed studies have included environmental and economic aspects together. These findings align with the observations presented by Janjua et al. [59], where the authors said that the majority of studies reviewed by them, almost 80.4%, discussed about environmental aspects, 18.6% addressed both the environmental as well as the economic aspects, only three studies discussed about social aspects while only five talked about all the three pillars of sustainability together.

The inferences from case studies contribute to a great extent to the information which is existing about LCA in the building sector. These results will act as a reference for engineers and designers which will help them in designing and constructing high-rise structures in a more sustainable as well as environmentally friendly way, as they will be able to understand the major factors that contribute to environmental impacts and take decisions accordingly. A structured method for evaluating the small-scale industrial building's construction was proposed so that eco-efficient solutions can be implemented in them [81]. Further, from the review of selected research articles, analyses of review articles and case studies, and review of literature about culture in LCA studies of buildings, it is observed that there are no such building LCA or LCSA studies, neither at the international nor at the national level, that have incorporated the concept of culture, the fourth pillar of sustainability in them. But researchers in other fields who have addressed culture in LCA or LCSA studies have said that there is definitely a scope for incorporating culture in LCA and LCSA studies, and culture should be presented distinctly from social aspects so that they do not lose their visibility, and this in turn might help in preserving the culture of the end users who have precise cultural needs and concerns [87].

Conclusions

The paper presents an elaborate review of the various types of LCA studies done till date by different researchers throughout the world, through a thorough review of research and review articles, and case studies. It is observed that most of the findings of the current study align with the findings of earlier review studies as discussed in the Discussion section. But one of the main and unique finding of the present study is that culture, the fourth pillar of sustainability has not been addressed in any of the LCA or LCSA studies in the building and construction industry. Past researchers have suggested to develop and strengthen social indicators, but none have talked about addressing culture, which forms the heart of the Sustainable Development Goals and is also deep-rooted in the lives of different groups of people all over the world. Inclusion of culture in LCA or LCSA might help in protecting the rich cultural diversity found in different communities worldwide. Further, a summary of the findings from the review of literature has been presented below which might act as future research directions for researchers in this field, and they are as follows:

i. A successful governing of the rehabilitation process of built heritage, by means of sustainability principles, can lead to a nexus of preservation of culture and sustainability.
ii. The development of social indicators has to be strengthened in the building sector for responding appropriately to the current challenges and coming up with a holistic picture.
iii. As problems are still there in dynamic data processing, so evaluation with present dynamic Building Information Modelling (BIM) integrated LCA tool should be integrated with Internet of things (IoT) big data information platform and the database should be broadened as well. Also, technical performance and application efficiency can be improved by integrating recent technologies like semantic web and Geographic information systems (GIS) technology.
iv. It should become mandatory to integrate LCA in the procurement process in the future for reducing carbon dioxide emissions.
v. Dynamic building life cycle assessment (LCA) can serve as an approach which is quite promising for improving the spatial as well as temporal accuracy of traditional LCA.
vi. A comprehensive framework needs to be developed to address the Triple Botom Line (TBL) aspects of sustainability throughout the entire life cycle of the buildings.
vii. Culture has a place in life cycle sustainability assessment (LCSA) and representing culture distinctively will have some potential benefits.
viii. None of the studies on LCA or LCSA in building and construction industry have addressed the concept of culture distinctively.
ix. Future studies, especially studies related to LCA or LCSA of buildings should include the concept of culture in them.

Author Contributions The article has been conceptualized by the main author (Manali Basu) after due discussion, and as per the suggestions of Nagaraju Kaja and Prashanti Rao respectively. The literature search and drafting has been performed by Manali Basu, in consultation with Nagaraju Kaja and Prashanti Rao. This study is a part of an ongoing PhD research work of Manali Basu at the School of Planning and Architecture, Vijayawada.

Funding The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Declarations

Conflict of interest The authors have not disclosed any competing interests.

References

1. H. Lim, S. Tae, S. Roh, Analysis of the primary building materials in support of G-SEED life cycle assessment in South Korea. Sustainability **10**(8), 2820 (2018). https://doi.org/10.3390/su10082820
2. A. Gravagnuolo, M. Angrisano, M. Nativo, Evaluation of environmental impacts of historic buildings conservation through life cycle assessment in a circular economy perspective. Aestimum (2021). https://doi.org/10.13128/aestim-10004
3. A.-F. Marique, B. Rossi, Cradle-to-grave life-cycle assessment within the built environment: comparison between the refurbishment and the complete reconstruction of an office building in Belgium. J. Environ. Manage. **224**, 396–405 (2018). https://doi.org/10.1016/j.jenvman.2018.02.055
4. K. Piparsania, P.C. Kalita, Development of DASH: design assessment framework for sustainable housing. Sustainability **14**(23), 15990 (2022). https://doi.org/10.3390/su142315990
5. N. Shandilya, A.R. Ghorpade, Handbook: ECBC compliance in Indian cities, ICLEI- local governments for sustainability, South Asia (ICLEI South Asia) (2019)
6. M. Jain, R. Rawal, Emissions from a net-zero building in India: life cycle assessment. Build. Cities **3**(1), 398–416 (2022). https://doi.org/10.5334/bc.194
7. M.M. Khasreen, P.F.G. Banfill, G.F. Menzies, Life-cycle assessment and the environmental impact of buildings: a review. Sustainability **1**(3), 674–701 (2009). https://doi.org/10.3390/su1030674
8. M. Tavana, M. Izadikhah, R.F. Saen, R. Zare, An integrated data envelopment analysis and life cycle assessment method for performance measurement in green construction management. Environ. Sci. Pollut. Res. **28**(1), 664–682 (2021). https://doi.org/10.1007/s11356-020-10353-7
9. A. Ciroth, M. Finkbeiner, J. Hildenbrand, W. Klöpffer, B. Mazijn, S. Prakash, G. Sonnemann, M. Traverso, C. M. L. Ugaya, S. Valdivia, G. Vickery-Niederman, Towards a life cycle sustainability assessment: making informed choices on products, United Nations Environment Programme (UNEP), (2011). https://www.unep.org/resources/report/towards-life-cycle-sustainability-assessment-making-informed-choices-products
10. Y. Dong, S.T. Ng, P. Liu, Towards the principles of life cycle sustainability assessment: an integrative review for the construction and building industry. Sustain. Cities Soc. **95**, 104604 (2023). https://doi.org/10.1016/j.scs.2023.104604
11. V.G. Larsen, N. Tollin, P.A. Sattrup, M. Birkved, T. Holmboe, What are the challenges in assessing circular economy for the built environment? A literature review on integrating LCA, LCC and S-LCA in life cycle sustainability assessment. LCSA. J. Build. Eng. **50**, 104203 (2022). https://doi.org/10.1016/j.jobe.2022.104203
12. J.G. Backes, M. Traverso, Application of life cycle sustainability assessment in the construction sector: a systematic literature review. Processes **9**(7), 1248 (2021). https://doi.org/10.3390/pr9071248
13. K. Nurse, Culture as the fourth pillar of sustainable development. Commonw. Secr. **11**, 28 (2006)
14. A.A. Rahim, S.N. Musa, S. Ramesh, M.K. Lim, A systematic review on material selection methods. Proc. Inst. Mech. Eng. L: J. Mater.: Des. Appl. (2020). https://doi.org/10.1177/1464420720916765
15. I. S. O. (ISO), ISO 14040:2006 Environmental management life cycle assessment principles and framework (2006).
16. T.E. McGrath, S.V. Nanukuttan, K. Owens, P.M. Basheer, P. Keig, Retrofit versus new-build house using life-cycle assessment. Proc. Inst. Civ. Eng. Eng. Sustain. **166**(3), 122–137 (2012). https://doi.org/10.1680/ensu.11.00026
17. T. Ekvall, Attributional and consequential life cycle assessment, in *Sustainability Assessment at the 21st century*. ed. by M.J. Bastante-Ceca, J.L. Fuentes-Bargues, L. Hufnagel, F.-C. Mihai, C. Iatu (IntechOpen, London, 2019). https://doi.org/10.5772/intechopen.89202
18. A. Atmaca, Life cycle assessment and cost analysis of residential buildings in south east of Turkey: part 1—review and methodology. Int. J. Life Cycle Assess. **21**(6), 831–846 (2016). https://doi.org/10.1007/s11367-016-1050-8
19. A. Atmaca, Life cycle assessment and cost analysis of residential buildings in south east of Turkey: part 2—a case study. Int. J. Life Cycle Assess. **21**(7), 925–942 (2016). https://doi.org/10.1007/s11367-016-1051-7
20. V. Pachta, V. Giourou, Comparative life cycle assessment of a historic and a modern school building, located in the city of Naoussa, Greece. Sustainability **14**(7), 4216 (2022). https://doi.org/10.3390/su14074216
21. M.W. Ryberg, P.K. Ohms, E. Møller, T. Lading, Comparative life cycle assessment of four buildings in Greenland. Build. Environ. **204**(15), 108130 (2021). https://doi.org/10.1016/j.buildenv.2021.108130
22. C.X. Chen, F. Pierobon, S. Jones, I. Maples, Y. Gong, I. Ganguly, Comparative life cycle assessment of mass timber and concrete residential buildings: a case study in China. Sustainability **14**(1), 144 (2021). https://doi.org/10.3390/su14010144
23. F. Jahandideh, S.N. Raman, M. Jamil, Z.I. Syed, Carbon footprint assessment in the life-cycle design of concrete structures in the tropics: a case study of residential buildings in Malaysia. J. Des. Built Environ. **20**(2), 27–34 (2020)
24. G.J. Lankhorst, J. Arts, K.C. Terwel, H.M. Jonkers, Life cycle analysis: load-bearing structures of high-rise buildings in Western Europe. CTBUH J. **3**, 28 (2019)
25. B. Petrovic, J.A. Myhren, X. Zhang, M. Wallhagen, O. Eriksson, Life cycle assessment of building materials for a single-family house in Sweden. Energy Proc. **158**, 3547–3552 (2019). https://doi.org/10.1016/j.egypro.2019.01.913
26. S. Roh, S. Tae, R. Kim, S. Park, Probabilistic analysis of major construction materials in the life cycle embodied environmental cost of Korean apartment buildings. Sustainability **11**(3), 846 (2019). https://doi.org/10.3390/su11030846
27. P. Vitale, U. Arena, An attributional life cycle assessment for an Italian residential multifamily building. Environ. Technol. **39**(23), 3033–3045 (2017). https://doi.org/10.1080/09593330.2017.1371252

28. V. Kumar, K. Hewage, R. Sadiq, Life cycle assessment of residential buildings: a case study in Canada. Int. J. Energy Environ. Eng. **9**(8), 1009–1017 (2015)
29. F. Asdrubali, C. Baldassarri, V. Fthenakis, Life cycle analysis in the construction sector: guiding the optimization of conventional Italian buildings. Energy Build. **64**, 73–89 (2013). https://doi.org/10.1016/j.enbuild.2013.04.018
30. B. Petrović, X. Zhang, O. Eriksson, M. Wallhagen, Life cycle cost analysis of a single-family house in Sweden. Buildings **11**(5), 215 (2021). https://doi.org/10.3390/buildings11050215
31. A. Noshadravan, T.R. Miller, J.R. Gregory, A lifecycle cost analysis of residential buildings including natural hazard risk. J. Constr. Eng. Manage. **143**(7), 04017017 (2017). https://doi.org/10.1061/(ASCE)CO.1943-7862.0001286
32. N.D. Lagaros, Life-cycle cost analysis of design practices for RC framed structures. Bull. Earthq. Eng. **5**(3), 425–442 (2007). https://doi.org/10.1007/s10518-007-9038-1
33. S. Vela, C. Calderini, P. Rosasco, C. Strazza, Economic and environmental evaluation of a single-story steel building in its life cycle: a comprehensive analysis. Sustainability **14**(21), 14638 (2022). https://doi.org/10.3390/su142114638
34. J.P. Carvalho, F.S. Villaschi, L. Bragança, Assessing life cycle environmental and economic impacts of building construction solutions with BIM. Sustainability **13**(16), 8914 (2021). https://doi.org/10.3390/su13168914
35. S. Liang, H. Gu, R. Bergman, Environmental life-cycle assessment and life-cycle cost analysis of a high-rise mass timber building: a case study in Pacific Northwestern United States. Sustainability **13**(14), 7831 (2021). https://doi.org/10.3390/su13147831
36. B. Kiss, J.D. Silvestre, R.A. Santos, Z. Szalay, Environmental and economic optimisation of buildings in Portugal and Hungary. Sustainability **13**(24), 13531 (2021). https://doi.org/10.3390/su132413531
37. R. Ramírez-Villegas, O. Eriksson, T. Olofsson, Combined environmental and economic assessment of energy efficiency measures in a multi-dwelling building. Energies **12**(13), 2484 (2019). https://doi.org/10.3390/en12132484
38. T.P. Obrecht, S. Jordan, A. Legat, M.R.M. Saade, A. Passer, An LCA methodolody for assessing the environmental impacts of building components before and after refurbishment. J. Clean. Prod. **327**, 129527 (2021). https://doi.org/10.1016/j.jclepro.2021.129527
39. K. Struhala, M. Ostrý, Life-cycle assessment of a rural terraced house: a struggle with sustainability of building renovations. Energies **14**(9), 2472 (2021). https://doi.org/10.3390/en14092472
40. M. Hu, Cost-effective options for the renovation of an existing education building toward the nearly net-zero energy goal—life-cycle cost analysis. Sustainability **11**(8), 2444 (2019). https://doi.org/10.3390/su11082444
41. H. Nydahl, S. Andersson, A.P. Åstrand, T. Olofsson, Environmental performance measures to assess building refurbishment from a life cycle perspective. Energies **12**(2), 299 (2019). https://doi.org/10.3390/en12020299
42. M. Österbring, É. Mata, L. Thuvander, H. Wallbaum, Explorative life-cycle assessment of renovating existing urban housing-stocks. Build. Environ. **165**, 106391 (2019). https://doi.org/10.1016/j.buildenv.2019.106391
43. R. Ramírez-Villegas, O. Eriksson, T. Olofsson, Life cycle assessment of building renovation measures–trade-off between building materials and energy. Energies **12**(3), 344 (2019). https://doi.org/10.3390/en12030344
44. M. Karoglou, S.S. Kyvelou, C. Boukouvalas, C. Theofani, A. Bakolas, M. Krokida, A. Moropoulou, Towards a preservation-sustainability nexus: applying LCA to reduce the environmental footprint of modern built heritage. Sustainability **11**(21), 6147 (2019). https://doi.org/10.3390/su11216147
45. Y. Ostermeyer, H. Wallbaum, F. Reuter, Multidimensional Pareto optimization as an approach for site-specific building refurbishment solutions applicable for life cycle sustainability assessment. Int. J. Life Cycle Assess. **18**(9), 1762–1779 (2013). https://doi.org/10.1007/s11367-013-0548-6
46. E.G. Dascalaki, P. Argiropoulou, C.A. Balaras, K.G. Droutsa, S. Kontoyiannidis, Analysis of the embodied energy of construction materials in the life cycle assessment of Hellenic residential buildings. Energy Build. **232**, 110651 (2021). https://doi.org/10.1016/j.enbuild.2020.110651
47. S. Liu, S. Qian, Evaluation of social life-cycle performance of buildings: theoretical framework and impact assessment approach. J. Clean. Prod. **213**, 792–807 (2018). https://doi.org/10.1016/j.jclepro.2018.12.200
48. S.A. Hosseinijou, S. Mansour, M.A. Shirazi, Social life cycle assessment for material selection: a case study of building materials. Int. J. Life Cycle Assess. **19**(3), 620–645 (2014). https://doi.org/10.1007/s11367-013-0658-1
49. A. Onososen, I. Musonda, M.M. Tjebane, Drivers of BIM-based life cycle sustainability assessment of buildings: an interpretive structural modelling approach. Sustainability. **14**(17), 11052 (2022). https://doi.org/10.3390/su141711052
50. A. Oladazimi, S. Mansour, S.A. Hosseinijou, M.H. Majdfaghihi, Sustainability identification of steel and concrete construction frames with respect to triple bottom line. Buildings **11**(11), 565 (2021). https://doi.org/10.3390/buildings11110565
51. A.T. Balasbaneh, A.K.B. Marsono, S.J. Khaleghi, Sustainability choice of different hybrid timber structure for low medium cost single-story residential building: environmental, economic and social assessment. J. Build. Eng. **20**, 235–247 (2018). https://doi.org/10.1016/j.jobe.2018.07.006
52. G. Fajilla, E. Borri, M.D. Simone, L.F. Cabeza, L. Bragança, Effect of climate change and occupant behaviour on the environmental impact of the heating and cooling systems of a real apartment. A parametric study through life cycle assessment. Energies **14**(24), 8356 (2021). https://doi.org/10.3390/en14248356
53. D.M. de Souza, M. Lafontaine, F. Charron-Doucet, X. Bengoa, B. Chappert, F. Duarte, L. Lima, Comparative life cycle assessment of ceramic versus concrete roof tiles in the Brazilian context. J. Clean. Prod. **89**, 165–173 (2015)
54. Z. Chen, L. Chen, X. Zhou, L. Huang, M. Sandanayake, P.-S. Yap, Recent technological advancements in BIM and LCA integration for sustainable construction: a review. Sustainability **16**(3), 1340 (2024). https://doi.org/10.3390/su16031340
55. A. Fnais, Y. Rezgui, I. Petri, T. Beach, J. Yeung, A. Ghoroghi, S. Kubicki, The application of life cycle assessment in buildings: challenges, and directions for future research. Int. J. Life Cycle Assess. **27**(6), 627–654 (2022). https://doi.org/10.1007/s11367-022-02058-5
56. M. Scherz, A.A. Wieser, A. Passer, H. Kreiner, Implementation of life cycle assessment (LCA) in the procurement process of buildings: a systematic literature review. Sustainability **14**(24), 16967 (2022). https://doi.org/10.3390/su142416967
57. N.B. Yildiz, H. Arslan, E. Yilmaz, Life cycle assessment of building materials: literature rewiew. Düzce Üniversitesi Bilim ve Teknoloji Dergisi **8**(2020), 210–219 (2020). https://doi.org/10.29130/dubited.572810
58. M.N. Nwodo, C.J. Anumba, A review of life cycle assessment of buildings using a systematic approach. Build. Environ. **162**, 106290 (2019). https://doi.org/10.1016/j.buildenv.2019.106290
59. S.Y. Janjua, P.K. Sarker, W.K. Biswas, A review of residential buildings' sustainability performance using a life cycle assessment approach. J. Sustain. Res. **1**, e190006 (2019). https://doi.org/10.20900/jsr20190006

60. B.S. Alotaibi, S.A. Khan, M.A. Abuhussain, N. Al-Tamimi, R. Elnaklah, M.A. Kamal, Life cycle assessment of embodied carbon and strategies for decarbonization of a high-rise residential building. Buildings **12**(8), 1203 (2022). https://doi.org/10.3390/buildings12081203
61. R. Kurian, K.S. Kulkarni, P.V. Ramani, C.S. Meena, A. Kumar, R. Cozzolino, Estimation of carbon footprint of residential building in warm humid climate of India through BIM. Energies **14**(14), 4237 (2021). https://doi.org/10.3390/en14144237
62. S. Ameen, Life cycle assessment of a construction phase carbon footprint of a building. Think India J. **22**(37), 198–207 (2019)
63. A. Ishaq, R.A. Khan, S. Meezab, Life cycle environmental assessment of an office and residential building in Northern India. Int. J. Innov. Technol. Explor. Eng. **8**(42), 114–121 (2019)
64. D. Husain, R. Prakash, Ecological footprint assessment and reduction of an academic building in Shahdol (India). Eur. J. Sustain. Dev. Res. **2**(4), 42 (2018). https://doi.org/10.20897/ejosdr/3910
65. S. Kumbhar, N. Kulkarni, A.B. Rao, B. Rao, Environmental life cycle assessment of traditional bricks in Western Maharashtra, India. Energy Proc. **54**, 260–269 (2014). https://doi.org/10.1016/j.egypro.2014.07.269
66. V.A. Sharma, V. Shree, H. Nautiyal, Life cycle environmental assessment of an educational building in Northern India: a case study. Sustain. Cities Soc. **4**, 22–28 (2012)
67. S.D. Alone, Life cycle analysis of a green building. Int. Res. J. Eng. Technol. **07**(10), 1063–1069 (2020)
68. A.D. Pawar, W.S. Marathe, Life cycle cost analysis of residential building by using local state guidelines of maintenance and repair cost. Int. Res. J. Eng. Technol. **07**(08), 1065–1068 (2020)
69. S. Kaur, T.S. Sidana, V.K. Paul, Life cycle costing of building Envelope. Int. J. Creat. Res. Thoughts **8**(6), 1793–1802 (2020)
70. S.A. Deore, H.P. Ambre, Life cycle cost analysis of a residential building. Int. Res. J. Eng. Technol. **06**(05), 7259–7262 (2019)
71. N.N. Kale, D. Joshi, R. Menon, Life cycle cost analysis of commercial buildings with energy efficient approach. Perspect. Sci. **8**, 452–454 (2016). https://doi.org/10.1016/j.pisc.2016.04.102
72. V.T.N. Ganugula, S. Dey, M.V.P. Akula, K. Balamuthu, Life cycle assessment for a suburban building located within the vicinity using Revit Architecture. J. Build. Pathol. Rehabil. **7**(1), 56 (2022). https://doi.org/10.1007/s41024-022-00199-6
73. R.T.R. Prakash, K.K. Shukla, Life cycle energy analysis of a residential building with different envelopes and climates in Indian context. Appl. Energy **89**(1), 193–202 (2012). https://doi.org/10.1016/j.apenergy.2011.05.054
74. A. Kumar, P. Singh, N.R. Kapoor, C.S. Meena, K. Jain, K.S. Kulkarni, R. Cozzolino, Ecological footprint of residential buildings in composite climate of India—a case study. Sustainability **13**(21), 11949 (2021). https://doi.org/10.3390/su132111949
75. R.C. Shrikhande, P. Desale, Life cycle Assessment of different walling materials and their comparative analysis. Int. J. Sci. Eng. Res. **11**(10), 32–43 (2020)
76. S. Srivastava, U.I. Raniga, S. Misra, A methodological framework for life cycle sustainability assessment of construction projects incorporating TBL and decoupling principles. Sustainability **14**(1), 197 (2021). https://doi.org/10.3390/su14010197
77. A. Yadav, A. Yadav, R. Agrawal, A. Jamwal, Life cycle assessment and life cycle energy analysis of buildings: a review. Manuf. Technol. Res. **14**(1–2), 26–34 (2021)
78. L. Ma, R. Azari, M. Elnimeiri, A building information modeling-based life cycle assessment of the embodied carbon and environmental impacts of high-rise building structures: a case study. Sustainability **16**(2), 569 (2024). https://doi.org/10.3390/su16020569
79. E.Ö. Kirilmaz, B. Metin, Life cycle assessment of internal wall panels: a case study of Sumerbank Kayseri textile factory restoration process. J. Arch. Sci. Appl. **8**(1), 345–354 (2023). https://doi.org/10.30785/mbud.1258158
80. A. Lachat, K. Mantalovas, T. Desbois, O. Yazoghli-Marzouk, A.-S. Colas, G. Di Mino, A. Feraille, From buildings' end of life to aggregate recycling under a circular economic perspective: a comparative life cycle assessment case study. Sustainability **13**(17), 9625 (2021). https://doi.org/10.3390/su13179625
81. M. Marrero, C. Rivero-Camacho, A. Martínez-Rocamora, M.D. Alba-Rodríguez, J. Solís-Guzmán, Life cycle assessment of industrial building construction and recovery potential. Case studies in Seville. Processes **10**(1), 76 (2021). https://doi.org/10.3390/pr10010076
82. D.M. Morsi, W.S. Ismaeel, A.E.A. El-Hamed, Life cycle assessment of a residential building in Egypt: a case study. IOP Conf. Ser. Mater. Sci. Eng. **974**, 012028 (2020)
83. S.M. Tulevech, D.J. Hage, S.K. Jorgensen, C.L. Guensler, R. Himmler, S.H. Gheewala, Life cycle assessment: a multi-scenario case study of a low-energy industrial building in Thailand. Energy Build. **168**, 191–200 (2018). https://doi.org/10.1016/j.enbuild.2018.03.011
84. J. Hong, G. Q. Shen, Y. Feng, Life cycle assessment of green buildings: a case study in China. In: International Conference on Construction and Real Estate Management (ICCREM). (2013).
85. W.O. Collinge, J.C. DeBlois, M.E. Sweriduk, A.E. Landis, A.K. Jones, L.A. Schaefer, M.M. Bilec, Measuring whole-building performance with dynamic LCA: a case study of a green university building. In: International Symposium on Life Cycle Assessment and Construction. France (2012).
86. F. Sabatini, Culture as fourth pillar of sustainable development: perspectives for integration, paradigms of action. Eur. J. Sustain. Dev. **8**(3), 31–40 (2019)
87. S. Pizzirani, S.J. McLaren, M.E. Forster, P. Pohatu, T.T.W. Porou, T.A. Warmenhoven, The distinctive recognition of culture within LCSA: realising the quadruple bottom line. Int. J. Life Cycle Assess. **23**(3), 663–682 (2018). https://doi.org/10.1007/s11367-016-1193-7
88. S. Pizzirani, S.J. McLaren, J.K. Seadon, Is there a place for culture in life cycle sustainability assessment? Int. J. Life Cycle Assess. **19**(6), 1316–1330 (2014). https://doi.org/10.1007/s11367-014-0722-5
89. T. Ketola, T. Salmi, Sustainability life cycle comparison of biofuels: sewage the saviour? Manag. Environ. Qual. **21**(6), 796–811 (2010). https://doi.org/10.1108/14777831011077655
90. B.P. Weidema, E. Lindeijer, Physical impacts of land use in product life cycle assessment. Final report of the EURENVIRONLCAGAPS sub-project on land use (2001).
91. B. Soust-Verdaguer, C. Llatas, A. García-Martínez, Simplification in life cycle assessment of single-family houses: a review of recent developments. Build. Environ. **103**, 215–227 (2016). https://doi.org/10.1016/j.buildenv.2016.04.014
92. A.F.A. Rashid, S. Yusoff, A review of life cycle assessment method for building industry. Renew. Sustain. Energy Rev. **45**, 244–248 (2015). https://doi.org/10.1016/j.rser.2015.01.043
93. A. Mastrucci, A. Marvuglia, U. Leopold, E. Benetto, Life cycle assessment of building stocks from urban to transnational scales: a review. Renew. Sustain. Energy Rev. **74**, 316–332 (2017). https://doi.org/10.1016/j.rser.2017.02.060
94. T.P. Obrecht, M. Röck, E. Hoxha, A. Passer, BIM and LCA integration: a systematic literature review. Sustainability **12**(14), 5534 (2020). https://doi.org/10.3390/su12145534

95. B. Soust-Verdaguer, C. Llatas, A. García-Martínez, Critical review of BIM-based LCA method to buildings. Energy Build. **136**, 110–120 (2017). https://doi.org/10.1016/j.enbuild.2016.12.009

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.